

# Panel MS-AR(1) after a common random-walk trend

Seasonally adjusted real GDP per worker

## 1 Model

Two-step. Step 1 builds a *common* trend from average growth of countries observed in consecutive quarters (not from the level mean, which falls when poorer countries enter). Log seasonally adjusted real GDP per worker, 2026-09-17 vintage, from 1987Q1 onward.

$$\Delta\tau_t = \frac{1}{n_t^{\text{cont}}} \sum_{i \in C_t \cap C_{t-1}} (y_{it} - y_{i,t-1}), \quad (1)$$

$$\tau_t = \tau_{t_0} + \sum_{s=t_0+1}^t \Delta\tau_s. \quad (2)$$

The level of  $\tau$  is pinned to the balanced-core mean at  $t_0$ . Mean  $\Delta\tau$  is 0.0045 per quarter (3 countries present in every quarter). Step 2 is estimated on  $y_{it}^* = y_{it} - \hat{\tau}_t$ , with no extra *gt*:

$$y_{it}^* = a_i + b_i \lambda^{t-T_{i0}} + z_{it}, \quad (3)$$

$$a_i \sim \mathcal{N}(\alpha, \omega^2), \quad b_i \sim \mathcal{N}(0, \omega_b^2) \text{ independent}, \quad (4)$$

$$z_{i,t+1} = (1 - \rho(s_{it}))\mu(s_{it}) + \rho(s_{it})z_{it} + \sigma(s_{it})\varepsilon_{it}. \quad (5)$$

$\lambda = 0.98$  is fixed (Barro 2% per year).  $|\rho| < 0.95$ . Three regimes; median  $\mu$  pinned at 0;  $\sigma$  and  $\rho$  switch.  $T_{i0}$  is country  $i$ 's first observation after filters. Random effects:  $7 \times 7$  Gauss–Hermite. Plotted cycles use posterior-mean  $a_i$  and  $b_i$ . The common trend  $\tau_t$  is discarded for the structural MS-AR. Standard errors ignore step-1 uncertainty.

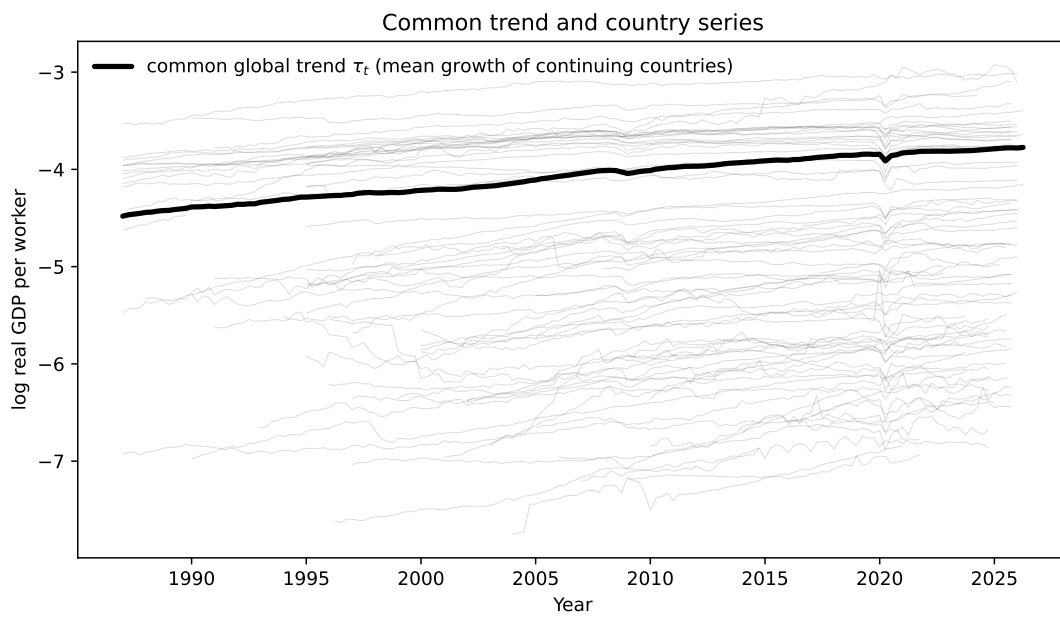


Figure 1: Faint lines: country log real GDP per worker. Thick line: common  $\tau_t$  from average growth of continuing countries (new entrants do not shift the level).

## 2 Common RW trend removed; RE intercepts plus Barro catch-up

2026-09-17 SA panel,  $y_{it} - \hat{\tau}_t$ , from 1987. 74 countries, 8291 observations, 1987-Q1–2026-Q2.  $|\rho| < 0.95$ ,  $\lambda = 0.98$  fixed. Fitted countries: 74. Observations: 8291. Log-likelihood: 22145.54. Ergodic mean of the cycle  $E[z] = 0.0465$  (0.0093). Permanent RE intercepts and catch-up  $y_{it} = a_i + b_i \lambda^{t-T_{i0}} + z_{it}$  with  $\alpha = -1.0050$  (0.0132),  $\omega = 0.4508$  (0.0023),  $\lambda = 0.9800$  (fixed),  $\omega_b = 0.2696$  (0.0024). Posterior-mean  $a_i$ : mean -1.173, min -2.696, max 0.686. Posterior-mean  $b_i$ : mean 0.015, min -1.011, max 1.011.

Table 1: Regime AR(1), transition matrix, and ergodic  $\pi$ . Columns are regimes.

|          | (0)                 | (1)                | (2)                |
|----------|---------------------|--------------------|--------------------|
| $\mu$    | -0.0326<br>(0.0507) | 0.0000<br>—        | 0.0947<br>(0.0148) |
| $\sigma$ | 0.0781<br>(0.0039)  | 0.0206<br>(0.0008) | 0.0081<br>(0.0002) |
| $\rho$   | 0.9498<br>(0.0010)  | 0.9500<br>(0.0000) | 0.9500<br>(0.0000) |
| $\Pi_0.$ | 0.6688<br>(0.0410)  | 0.3231<br>(0.0451) | 0.0081<br>(0.0142) |
| $\Pi_1.$ | 0.0471<br>(0.0076)  | 0.8991<br>(0.0102) | 0.0537<br>(0.0072) |
| $\Pi_2.$ | 0.0081<br>(0.0029)  | 0.0361<br>(0.0054) | 0.9558<br>(0.0046) |
| $\pi$    | 0.0713<br>(0.0097)  | 0.4130<br>(0.0238) | 0.5157<br>(0.0275) |

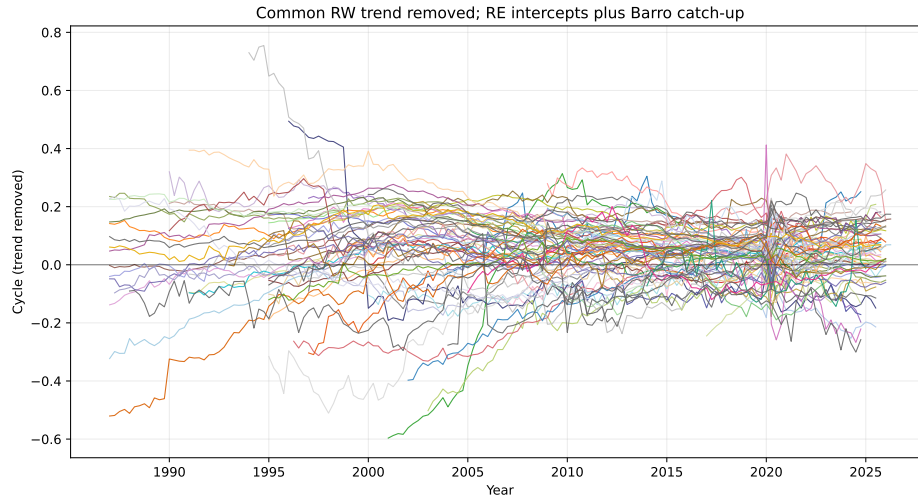


Figure 2: Country cycles after removing  $\tau_t$ ,  $a_i$ , and  $b_i \lambda^{t-T_{i0}}$ .